Hints: Polynomials – 7. Interpolation and finite differences

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Write f in Newton's form $f(x) = \sum_k \Delta^k f(0)\binom{x}{k}$ (both sides have degree $\leqslant \deg f$ and agree at $0, 1, \ldots, \deg f$). If f is integer-valued, each $\Delta^k f(0)$ is an integer combination of the values $f(0), \ldots, f(k)$.

Hint 2. Examples: $\binom{x}{n} = \frac{x(x-1)\cdots(x-n+1)}{n!}$ is integer-valued of degree n with leading coefficient $1/n! \notin \mathbb{Z}$ for every $n \geqslant 2$; e.g. $\binom{x}{2} = \frac{x^2}{2} - \frac{x}{2}$. (The constant term $f(0)$ is always an integer.)

Problem 2

Hint 1. Put $m = \lfloor n/k \rfloor$ and $r = n - km \in \{0, \ldots, k-1\}$. Find a polynomial identity in n and m that holds because r is one of $0, \ldots, k-1$.

Hint 2. $\prod_{j=0}^{k-1}(n - km - j) = 0$, since one factor is $r - r$. Expanded in powers of m, this is $(-k)^k m^k + \sum_{i<k} Q_i(n) m^i = 0$ with polynomials Q_i; divide by $-(-k)^k$.

Problem 3

Hint 1. Write $g_n = \sum_j c_{n,j} x^j$. The rule says $c_{n+1,j} = c_{n,j-1} + c_{n,j+1}$, except that nothing is carried to exponent -1. Interpret $c_{n,j}$ as a number of lattice paths.

Hint 2. $c_{n,j}$ counts walks of n steps ± 1 from 0 to j that never go below 0. So $g_n(0)$ is the number of Dyck paths of length n: 0 for odd n and the Catalan number $C_{n/2}$ for even n.

Hint 3. Answer: $g_{15}(0) = 0$ and $g_{16}(0) = C_8 = \frac{1}{9}\binom{16}{8} = 1430$.

Problem 4

Hint 1. All differences of 2^k equal 2^k again, so $\Delta^j(2^k)\big|_{k=0} = 1$. Write f in Newton's form.

Hint 2. $f(x) = \sum_{j=0}^{n} \binom{x}{j}$. Then $f(n+1) = \sum_{j \leqslant n} \binom{n+1}{j} = 2^{n+1} - 1$ and $f(n+2) = \sum_{j \leqslant n} \binom{n+2}{j} = 2^{n+2} - n - 3$. In terms of differences: f agrees with 2^k except that its $(n+1)$-st difference is 0 instead of 1.

Problem 5

Hint 1. Bound $\max_{[0,1]} |P|$ from below by an integral that is a positive rational number with controlled denominator. Try $\int_0^1 P(x)^{2m}\,dx$ for large m.

Hint 2. $\int_0^1 P^{2m}$ is positive and a sum of terms (integer)$/(j+1)$ with $j \leqslant 2mn$, hence at least $1/\operatorname{lcm}(1, \ldots, 2mn+1) = e^{-\psi(2mn+1)}$, where ψ is Chebyshev's function. So $\max_{[0,1]} |P| \geqslant \exp(-\psi(2mn+1)/(2m))$ for every m.

Hint 3. Let $m \to \infty$: since $\psi(N)/N \to 1$ (prime number theorem; in fact $\liminf \psi(N)/N \leqslant 1$ suffices, which follows from Mertens' estimate $\sum_{j \leqslant N} \Lambda(j)/j = \ln N + O(1)$), the right-hand side gets arbitrarily close to e^{-n} along a subsequence, hence $\max |P| \geqslant e^{-n}$.

Problem 6

Hint 1. Forward difference formula: $f(x) = \sum_k \Delta^k f(0)\binom{x}{k}$, proved by checking both sides agree at $x = 0, \dots, \deg f$ (or by $E^x = (I + \Delta)^x$ on integers). Combine with $\sum_{k=0}^{m} \binom{k}{j} = \binom{m+1}{j+1}$.

Hint 2. $k^3 = 6\binom{k}{3} + 6\binom{k}{2} + \binom{k}{1}$ (differences of k^3 at 0: $0, 1, 6, 6$), so $\sum_{k=1}^{m} k^3 = 6\binom{m+1}{4} + 6\binom{m+1}{3} + \binom{m+1}{2} = \left(\frac{m(m+1)}{2}\right)^2$. Also $k(k+1)(k+2) = 6\binom{k+2}{3}$, so the second sum is $6\binom{m+3}{4} = \frac{m(m+1)(m+2)(m+3)}{4}$.

Problem 7

Hint 1. $P(x) - 7 = (x - a_1)(x - a_2)(x - a_3)(x - a_4)Q(x)$ with $Q \in \mathbb{Z}[x]$.

Hint 2. If $P(k) = 14$, then $k - a_1, \dots, k - a_4$ are four distinct integers whose product divides 7. The only divisors of 7 are $\pm 1, \pm 7$, and the product of four distinct ones is 49, which does not divide 7.

Hint 3. General statement: if $P \in \mathbb{Z}[x]$ takes the value c at four (or more) distinct integers, then $P(k) - c$ is never ± 1 or $\pm$ a prime (more generally, $P(k) - c$ is divisible by a product of four distinct nonzero integers).

Problem 8

Hint 1. Vandermonde: $\det(x_i^j)$ is a polynomial in x_n of degree n (say $n + 1$ nodes $x_0, \dots, x_n$) vanishing at $x_0, \dots, x_{n-1}$, so it equals the smaller Vandermonde determinant times $\prod_{i<n}(x_n - x_i)$; induct.

Hint 2. The coefficient vector is $V^{-1}y$. By Cramer's rule each coefficient is $\det V_k / \det V$, where V_k has entries in $[0, 1]$ except one column equal to y, so $|\det V_k| \leqslant (n+1)! \max |y_i|$, while $|\det V| = \prod_{i<j} |x_j - x_i| \geqslant \delta^{n(n+1)/2}$. So every coefficient is at most $(n+1)!\, \delta^{-n(n+1)/2} \max |y_i|$. (Some dependence on δ is unavoidable: the nodes $0, \delta$ with values $0, 1$ give x/δ.)

Problem 9

Hint 1. By Lagrange, $|P(x)| \leqslant \sum_{k=0}^{n} |L_k(x)|$ with $L_k(x) = \prod_{j \neq k} \frac{x-j}{k-j}$ and $\prod_{j \neq k} |k - j| = k!\,(n-k)!$. It suffices to show $\prod_{j \neq k} |x - j| \leqslant n!$ for $x \in [0, n]$; then the sum is at most $\sum_k \binom{n}{k} = 2^n$.

Hint 2. Let $x \in [m, m+1]$, $0 \leqslant m \leqslant n-1$, and by symmetry $k \leqslant m$. Bound $|x - j| \leqslant j - m$ for $j > m$ and $|x - j| \leqslant m + 1 - j$ for $j \leqslant m$: the product is at most $(n-m)!\,(m+1)!/(m+1-k) \leqslant (m+1)!\,(n-m)!$, and this is $\leqslant n!$ because $\binom{n}{m} \geqslant m + 1$.

Hint 3. Sharpness: take $P(k) = (-1)^k$. Then $\Delta^n P = (-2)^n$, so the leading coefficient is $(-2)^n/n!$, and by Chebyshev a polynomial with leading coefficient a has $\sup_{[0,n]} |P| \geqslant 2|a|(n/4)^n$. This gives $\sup |P| \geqslant 2(n/2)^n/n!$, which grows like $(e/2)^n/\sqrt{n}$.

Problem 10

Hint 1. Define $P[x_0, \dots, x_k]$ as the leading coefficient (of x^k) of the interpolating polynomial L_k through $(x_i, P(x_i))$, $i \leqslant k$. Reading off the x^k coefficient of Lagrange's formula gives the stated symmetric formula; check it satisfies the recursion $P[x_0, \dots, x_k] = \frac{P[x_1, \dots, x_k] - P[x_0, \dots, x_{k-1}]}{x_k - x_0}$.

Hint 2. Newton: $L_k - L_{k-1}$ has degree $\leqslant k$ and vanishes at $x_0, \dots, x_{k-1}$, so it equals $P[x_0, \dots, x_k] \prod_{j<k}(x - x_j)$. Summing, $L_n(x) = \sum_{k \leqslant n} P[x_0, \dots, x_k] \prod_{j<k}(x - x_j)$.

Hint 3. If $\deg P = n$, the interpolant through $n + 1$ nodes is P itself, so its order-n divided difference is the leading coefficient of P.